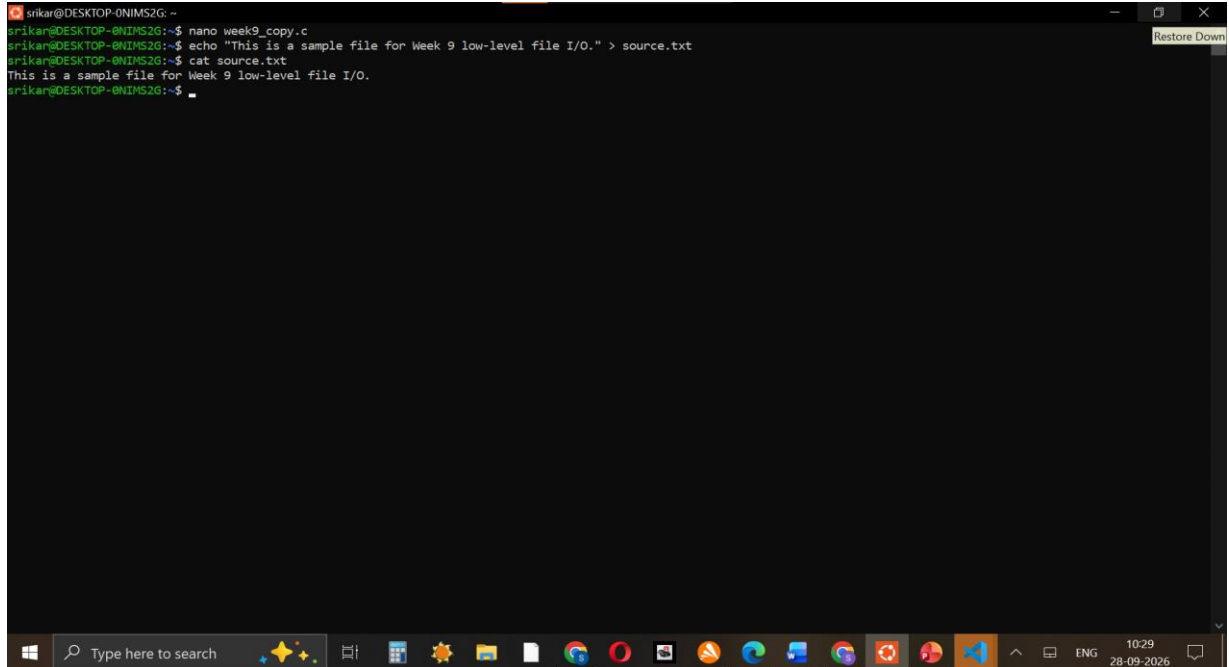


Practical Week-9

1. Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite).



```
srikar@DESKTOP-0NIMS2G: ~  
srikar@DESKTOP-0NIMS2G:~$ nano week9_copy.c  
srikar@DESKTOP-0NIMS2G:~$ echo "This is a sample file for Week 9 low-level file I/O." > source.txt  
srikar@DESKTOP-0NIMS2G:~$ cat source.txt  
This is a sample file for Week 9 low-level file I/O.  
srikar@DESKTOP-0NIMS2G:~$
```

The screenshot shows a Windows terminal window with a dark background. The command prompt is at the top left, showing the user 'srikar' and the machine 'DESKTOP-0NIMS2G'. The terminal displays the execution of a file copy utility. The user runs 'nano week9_copy.c' to create a source file. Then, they run 'echo "This is a sample file for Week 9 low-level file I/O." > source.txt' to create a source file. Finally, they run 'cat source.txt' to display the contents of the source file, which is 'This is a sample file for Week 9 low-level file I/O.'. The terminal window has a standard Windows taskbar at the bottom with various application icons and a system tray showing the time as 10:29 on 28-09-2026.

Output:

```
srikan@DESKTOP-0NIMS2G: ~  
srikan@DESKTOP-0NIMS2G:~$ nano week9_copy.c  
srikan@DESKTOP-0NIMS2G:~$ echo "This is a sample file for Week 9 low-level file I/O." > source.txt  
srikan@DESKTOP-0NIMS2G:~$ cat source.txt  
This is a sample file for Week 9 low-level file I/O.  
srikan@DESKTOP-0NIMS2G:~$ gcc week9_copy.c -o week9_copy  
srikan@DESKTOP-0NIMS2G:~$ ./week9_copy  
File copied successfully using low-level system calls.  
Current file position: 53 bytes  
srikan@DESKTOP-0NIMS2G:~$
```



```

#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main(int argc, char *argv[]) {
    int source, destination;
    char buffer[1024];
    ssize_t bytes;
    off_t size;

    if (argc != 3) {
        printf("Usage: %s source destination\n", argv[0]);
        return 1;
    }

    source = open(argv[1], O_RDONLY);
    if (source < 0) {
        perror("Error opening source file");
        return 1;
    }

    destination = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (destination < 0) {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    // Get source file size using lseek
    size = lseek(source, 0, SEEK_END);
    printf("Source file size: %ld bytes\n", (long)size);
    lseek(source, 0, SEEK_SET); // Reset file offset back to start

    // Copy file content
    while ((bytes = read(source, buffer, sizeof(buffer))) > 0) {
        write(destination, buffer, bytes);
    }

    close(source);
    close(destination);

    printf("File copied successfully using system calls.\n");
    return 0;
}

```

Modified

program to use lseek():

1B

```
srikar@DESKTOP-0NIMS2G: ~  
This is a sample file for Week 9 low-level file I/O.  
srikar@DESKTOP-0NIMS2G:~$ nano week9_standard.c  
srikar@DESKTOP-0NIMS2G:~$ gcc week9_standard.c -o week9_standard  
srikar@DESKTOP-0NIMS2G:~$ ./week9_standard  
File copied successfully using standard library functions.  
srikar@DESKTOP-0NIMS2G:~$ cat copy_standard.txt  
This is a sample file for Week 9 low-level file I/O.  
srikar@DESKTOP-0NIMS2G:~$ time ./week9_copy  
File copied successfully using low-level system calls.  
Current file position: 53 bytes  
  
real    0m0.006s  
user    0m0.000s  
sys     0m0.006s  
srikar@DESKTOP-0NIMS2G:~$ time ./week9_standard  
File copied successfully using standard library functions.  
  
real    0m0.002s  
user    0m0.000s  
sys     0m0.002s  
srikar@DESKTOP-0NIMS2G:~$ nano week9_redirect.c  
srikar@DESKTOP-0NIMS2G:~$  
srikar@DESKTOP-0NIMS2G:~$ echo "This is input redirection using dup2 in Linux." > input.txt  
srikar@DESKTOP-0NIMS2G:~$ gcc week9_redirect.c -o week9_redirect  
srikar@DESKTOP-0NIMS2G:~$ ./week9_redirect  
srikar@DESKTOP-0NIMS2G:~$ cat output.txt  
This is input redirection using dup2 in Linux.  
srikar@DESKTOP-0NIMS2G:~$
```

Output:

```
srikan@DESKTOP-0NIMS2G: ~  
srikan@DESKTOP-0NIMS2G:~$ nano week9_copy.c  
srikan@DESKTOP-0NIMS2G:~$ echo "This is a sample file for Week 9 low-level file I/O." > source.txt  
srikan@DESKTOP-0NIMS2G:~$ cat source.txt  
This is a sample file for Week 9 low-level file I/O.  
srikan@DESKTOP-0NIMS2G:~$ gcc week9_copy.c -o week9_copy  
srikan@DESKTOP-0NIMS2G:~$ ./week9_copy  
File copied successfully using low-level system calls.  
Current file position: 53 bytes  
srikan@DESKTOP-0NIMS2G:~$
```

2. Develop a program that redirects standard input and output using file descriptors and dup2(). Demonstrate how shells implement I/O redirection.

```
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
#include <stdlib.h>  
  
int main(void) {  
    int input_fd, output_fd;  
  
    input_fd = open("input.txt", O_RDONLY);  
    if (input_fd < 0) {  
        perror("input.txt");  
        return 1;  
    }  
  
    output_fd = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);  
    if (output_fd < 0) {  
        perror("output.txt");  
        close(input_fd);  
        return 1;  
    }  
  
    // Redirect standard input and standard output  
    dup2(input_fd, STDIN_FILENO);  
    dup2(output_fd, STDOUT_FILENO);  
  
    close(input_fd);  
    close(output_fd);  
  
    char buffer[100];  
    printf("This output is redirected to output.txt\n");  
    printf("Reading from redirected input:\n");  
  
    while (read(STDIN_FILENO, buffer, sizeof(buffer) - 1) > 0) {  
        printf("%s", buffer);  
    }  
  
    return 0;  
}
```

Output:

```
srikar@DESKTOP-0NIMS2G: ~  
srikar@DESKTOP-0NIMS2G:~$ nano week9_copy.c  
srikar@DESKTOP-0NIMS2G:~$ echo "This is a sample file for Week 9 low-level file I/O." > source.txt  
srikar@DESKTOP-0NIMS2G:~$ cat source.txt  
This is a sample file for Week 9 low-level file I/O.  
srikar@DESKTOP-0NIMS2G:~$ gcc week9_copy.c -o week9_copy  
srikar@DESKTOP-0NIMS2G:~$ ./week9_copy  
File copied successfully using low-level system calls.  
Current file position: 53 bytes  
srikar@DESKTOP-0NIMS2G:~$ cat copy_lowlevel.txt  
This is a sample file for Week 9 low-level file I/O.  
srikar@DESKTOP-0NIMS2G:~$ nano week9_standard.c  
srikar@DESKTOP-0NIMS2G:~$ gcc week9_standard.c -o week9_standard  
srikar@DESKTOP-0NIMS2G:~$ ./week9_standard  
File copied successfully using standard library functions.  
srikar@DESKTOP-0NIMS2G:~$ cat copy_standard.txt  
This is a sample file for Week 9 low-level file I/O.  
srikar@DESKTOP-0NIMS2G:~$
```